

⁹Be(³⁷Ca,X) 2024Dr01

$J^\pi=3/2^+$ for ³⁷Ca ground state.
2024Dr01: a 72-MeV/nucleon ³⁷Ca secondary beam was produced via the fragmentation of a 140-MeV/nucleon ⁴⁰Ca²⁰⁺ primary beam impinging on a 0.5-mm-thick Be target. Experimental setup includes the CAESium-iodide scintillator ARray (CAESAR) for detecting γ rays, a DSSD-CsI(Tl) ΔE -E Ring Telescope for detecting protons, a Scintillating-Fiber Array (SFA) and the S800 spectrograph for detecting heavy residuals. Measured E_p , E_γ , and $E(\text{residual})$. Deduced the first excited state in ³⁵Ca via the 2p+³³Ar exit channel from total decay-energy spectra using the invariant-mass method. Comparisons with shell-model calculations with the USDC Hamiltonian.

³⁵Ca Levels

<u>E(level)</u>	<u>J^{π}</u>	<u>Comments</u>
2.08×10 ³ 10	(3/2 ⁺)	E(level): From total decay energy of 2p+ ³³ Ar E _T =1667 keV 20. J ^{π} : shell-model calculations.